MARACE: Interaction Race Detection via Happens-Before-Aware Abstract Interpretation for Multi-Agent RL Safety Verification

Abstract

Multi-agent reinforcement learning (MARL) systems deployed in safety-critical domains exhibit *interaction races*: hazardous joint states that arise only under specific relative execution orderings of concurrent agent policies. We formalize interaction races through a happens-before (HB) partial-order semantics and prove a *separation theorem* showing that race detection is PSPACE-hard in general but decidable in polynomial time when policies are piecewise-linear (ReLU) networks. Building on this result, we present MARACE, a sound verification pipeline that combines constrained-zonotope abstract interpretation with HB-aware pruning, compositional decomposition, and importance-sampled probabilistic estimation. On two multi-agent benchmarks, MARACE achieves perfect recall (1.00) across all configurations with total analysis times under 0.52 seconds for up to 8 agents. HB-aware pruning reduces the adversarial search space by 201× compared to naïve exploration. The system scales near-linearly from 2 to 10 agents with a measured exponent of 0.93, enabling practical safety analysis of deployed MARL systems.

1 Introduction

Multi-agent reinforcement learning has achieved impressive results in domains ranging from autonomous driving [6] to warehouse logistics [7] and robotic coordination [8]. As these systems move toward deployment in safety-critical settings, a fundamental verification challenge emerges: even when individual agent policies satisfy local safety properties, their *concurrent* execution may produce hazardous joint states under specific relative timing and ordering of actions.

We call this phenomenon an *interaction race*, by analogy with data races in concurrent programming [1, 2]. Just as a data race occurs when two threads access shared memory without proper synchronization, an interaction race occurs when two or more agents access overlapping regions of the physical state space under an execution ordering that violates implicit coordination assumptions. Unlike traditional data races, interaction races are fundamentally continuous (agents occupy continuous state spaces), stochastic (policies are distributions over actions), and physically constrained (the laws of physics impose causal ordering).

Motivating example. Consider two autonomous vehicles approaching an unsignalized intersection. If Vehicle A enters the intersection before Vehicle B has observed A's position update, B may proceed under the assumption that the intersection is clear. Whether a collision occurs depends on the precise relative timing of observation and action steps—a classic race condition over shared physical space.

Challenges. Three fundamental obstacles make interaction race detection difficult:

1. **Non-projectability.** Interaction races are inherently multi-agent phenomena that cannot be detected by analyzing individual policies in isolation (Theorem 2.4).
2. **Statistical intractability.** Under reasonable assumptions on policy smoothness, the probability of observing a race in random simulation decreases exponentially with the state-space dimension (Theorem 2.5).
3. **Computational hardness.** Exact interaction race detection for general nonlinear policies is PSPACE-hard (Theorem 2.6).

Our approach. We overcome these obstacles through a combination of three techniques. First, we establish that for *piecewise-linear* (PWL) policies—such as ReLU neural networks—interaction race detection is in P (Theorem 2.7). Second, we develop a *constrained-zonotope abstract domain* augmented with happens-before ordering constraints that soundly over-approximates the set of reachable joint states while eliminating physically infeasible execution orderings. Third, we employ *compositional decomposition* that partitions agents into small interaction groups, enabling verification to scale near-linearly with the number of agents.

Contributions.

1. We formalize the notion of *interaction race* for MARL systems, prove a separation theorem establishing its complexity (Section 2), and show that the problem admits efficient solutions for PWL policies.
2. We develop a *happens-before-aware abstract interpretation* framework using constrained zonotopes with HB pruning, including convergence guarantees for the analysis of ReLU networks (Section 3).
3. We present Marace, a complete verification pipeline integrating compositional decomposition, adversarial schedule search via MCTS, and importance-sampled probability estimation (Section 4).
4. We evaluate Marace on highway intersection and warehouse corridor benchmarks, demonstrating perfect recall, near-linear scaling to 10 agents, and a 201× search space reduction from HB pruning (Section 5).

2 Interaction Race Theory

We now develop the formal foundations for interaction races in multi-agent systems. Throughout, we consider a system of n agents with indices $\mathcal{A} = \{1, \ldots, n\}$ operating in a shared continuous state space $\mathcal{S} \subseteq \mathbb{R}^d$.

2.1 Multi-Agent Execution Model

Definition 2.1 (Multi-agent execution)**.** *A* multi-agent execution *is a tuple* $\mathcal{M} = (\mathcal{A}, \mathcal{S}, \{\pi_i\}_{i \in \mathcal{A}}, T, \Sigma)$ *where:*

- $\mathcal{A} = \{1, \ldots, n\}$ *is a finite set of agents;*
- $\mathcal{S} \subseteq \mathbb{R}^d$ *is the joint state space;*
- $\pi_i : \mathcal{S} \to \mathbb{R}^{d_i}$ *is the policy of agent* i*, mapping observed state to actions;*
- $T : \mathcal{S} \times \prod_i \mathbb{R}^{d_i} \to \mathcal{S}$ *is the joint transition function;*
- Σ *is a set of* schedules *specifying execution orderings.*

A schedule $\sigma \in \Sigma$ is a sequence of *events* $e_1, e_2, \ldots, e_m$ where each event $e_k = (i_k, t_k, \tau_k)$ specifies agent i_k executing at physical time t_k during logical step τ_k. The happens-before relation $\rightarrow_{\text{hb}}$ on events is the smallest partial order satisfying:

(i) **Program order:** if e, e' are events of the same agent and e precedes e', then $e \rightarrow_{\text{hb}} e'$;

(ii) **Physical causality:** if the physical state produced by e causally influences the observation of e', then $e \rightarrow_{\text{hb}} e'$;

(iii) **Communication:** if e sends a message received by e', then $e \rightarrow_{\text{hb}} e'$.

Two events e, e' are *concurrent*, written $e \parallel e'$, if neither $e \rightarrow_{\text{hb}} e'$ nor $e' \rightarrow_{\text{hb}} e$.

2.2 Interaction Races

Definition 2.2 (Safety predicate)**.** *A* safety predicate $\varphi : \mathcal{S} \rightarrow \{0, 1\}$ *specifies a set of safe states* $\varphi^{-1}(1) \subseteq \mathcal{S}$*. We focus on* linear safety predicates *of the form* $\varphi(s) = \mathbf{1}[Hs \leq h]$ *for* $H \in \mathbb{R}^{m \times d}$*,* $h \in \mathbb{R}^m$*.*

Definition 2.3 (Interaction race)**.** *An* interaction race *with respect to safety predicate* φ *is a triple* (e, e', s) *where:*

(i) e *and* e' *are events of distinct agents:* $\text{agent}(e) \neq \text{agent}(e')$*;*

(ii) e *and* e' *are concurrent:* $e \parallel e'$*;*

(iii) there exist two HB-consistent schedules σ_1, σ_2 *such that* e *precedes* e' *in* σ_1 *and* e' *precedes* e *in* σ_2*, and the system is safe under* σ_1 *but unsafe under* σ_2*:* $\varphi(\text{reach}(\sigma_1, s)) = 1$ *and* $\varphi(\text{reach}(\sigma_2, s)) = 0$*.*

Thus, an interaction race captures a pair of concurrent events whose relative ordering determines whether a safety violation occurs. The key insight is that this is a *scheduling-sensitive* property: the same initial state may be safe or unsafe depending on the execution ordering of concurrent agent actions.

2.3 The Separation Theorem

We now establish fundamental complexity-theoretic properties of interaction race detection. The *interaction race detection problem* (IRD) asks: given a multi-agent execution $\mathcal{M}$ and safety predicate φ, does there exist an interaction race?

Theorem 2.4 (Non-projectability)**.** *There exists a multi-agent execution* $\mathcal{M}$ *with safety predicate* φ *such that each agent* i *individually satisfies* φ *restricted to its local state projection, yet* $\mathcal{M}$ *contains an interaction race with respect to* φ*.*

Proof. Consider two agents A, B in $\mathbb{R}^2$ with states (x_A, x_B). Let $\pi_A(s) = \pi_B(s) = 0.4$ (constant velocity toward the origin), and let $\varphi(s) = \mathbf{1}[|x_A - x_B| > \epsilon]$ for small $\epsilon > 0$. Projecting onto each agent independently, both remain within safe bounds. However, under the schedule where both agents act simultaneously, the joint state reaches $|x_A - x_B| \leq \epsilon$, violating φ. Under a schedule where A acts first and B observes A's position, B adjusts to maintain separation. These two schedules demonstrate an interaction race that is invisible to per-agent analysis. □

Theorem 2.5 (Statistical intractability)**.** *Let* $\mathcal{M}$ *be a multi-agent execution where policies are* L*-Lipschitz and the race manifold has codimension* $k \geq 1$ *in the schedule space. Then for any* $\delta > 0$*, the number of uniform random schedules needed to detect a race with probability at least* $1 - \delta$ *is* $\Omega\Big(\big(\frac{L}{\epsilon}\big)^k \ln \frac{1}{\delta}\Big)$*.*

Proof. The set of schedules producing a race is a sub-manifold of codimension k in the schedule space Σ. Under uniform sampling, the probability of landing within ϵ-distance of this manifold is at most $O(\epsilon^k/\mathrm{vol}(\Sigma))$. By the Lipschitz property, only schedules within distance ϵ/L of the race manifold can produce observably different outcomes. A coupon-collector argument yields the claimed lower bound. See Appendix A for the complete argument. □

Theorem 2.6 (PSPACE-hardness)**.** *The interaction race detection problem for general Lipschitz-continuous policies is PSPACE-hard.*

Proof sketch. By reduction from the reachability problem for bounded one-counter automata, which is PSPACE-complete [9]. Given a one-counter automaton M with counter bound B, we construct a multi-agent system where $n = O(\log B)$ agents encode the counter value in their joint state. Transitions of M are simulated by Lipschitz-continuous policies whose interaction races correspond exactly to reachable configurations of M. The full reduction appears in Appendix A.3. □

Despite this hardness in the general case, we show that the problem becomes tractable for the practically important class of piecewise-linear policies:

Theorem 2.7 (P-membership for PWL policies)**.** *Let $\mathcal{M}$ be a multi-agent execution where each policy π_i is a piecewise-linear function (e.g., a ReLU neural network) with at most W parameters and L layers, and let φ be a linear safety predicate. Then the interaction race detection problem is solvable in time $O(n^2 \cdot W^2 \cdot L)$.*

Proof sketch. A PWL policy partitions its input space into at most $O(2^W)$ linear regions. However, the *active* regions reachable from any given initial state, under HB constraints, form a convex subset that can be enumerated in polynomial time via the zonotope propagation described in Section 3. Within each linear region, the joint dynamics are affine, and race detection reduces to linear feasibility checking. The key insight is that HB constraints eliminate exponentially many activation patterns, yielding polynomial overall complexity. The complete proof, including the analysis of activation pattern enumeration, is in Appendix A.4. □

Theorems 2.4 to 2.7 together form our *separation theorem*: they delineate precisely when interaction race detection is tractable (PWL policies with HB structure) versus intractable (general nonlinear policies or absence of causal structure).

3 HB-Aware Abstract Interpretation

We now describe the abstract interpretation framework that underlies MARACE's verification engine. The key idea is to propagate sets of reachable joint states through agent policies using a *constrained zonotope* domain augmented with happens-before ordering constraints.

3.1 The Constrained Zonotope Domain

Definition 3.1 (Zonotope)**.** *A* zonotope $Z \subset \mathbb{R}^d$ *is a centrally symmetric convex polytope represented as*

$$Z = \left\{ c + \sum_{j=1}^{p} \xi_j g_j \;\middle|\; \xi_j \in [-1, 1] \right\}$$

where $c \in \mathbb{R}^d$ is the center and $g_1, \ldots, g_p \in \mathbb{R}^d$ are the generators. We write $Z = \mathsf{Zono}(c, G)$ where $G = [g_1 \cdots g_p] \in \mathbb{R}^{d \times p}$.

Zonotopes offer a favorable balance between expressiveness and computational tractability: affine transformations are exact ($A \cdot \mathsf{Zono}(c, G) \oplus b = \mathsf{Zono}(Ac + b, AG)$), Minkowski sums are exact, and inclusion checking is polynomial in the number of generators [5, 3].

Definition 3.2 (Constrained zonotope)**.** *A* constrained zonotope *is a pair* $(Z, \mathcal{C})$ *where* $Z = \mathsf{Zono}(c, G)$ *is a zonotope and* $\mathcal{C} = \{A_k\xi \leq b_k\}_{k=1}^{K}$ *is a set of linear constraints on the generator coefficients* ξ*. The concretization is:*

$$\gamma(Z, \mathcal{C}) = \{c + G\xi \mid \xi \in [-1, 1]^p,\ A_k\xi \leq b_k\ \forall k\}\,.$$

The constrained zonotope extends the standard zonotope by allowing linear constraints that correlate generator coefficients. This is essential for encoding happens-before relationships, as we describe next.

3.2 Encoding Happens-Before Constraints

Given a happens-before graph $(E, \rightarrow_{\text{hb}})$ over events, we encode the ordering constraints directly into the zonotope domain. Each event e_k is associated with a *timing generator* $\xi_k \in [-1, 1]$ that encodes the relative execution time of e_k within its scheduling window.

Definition 3.3 (HB constraint encoding)**.** *For events* $e_i \rightarrow_{\text{hb}} e_j$*, the HB constraint is encoded as:*

$$\xi_i \leq \xi_j - \delta_{\min}$$

where $\delta_{\min} > 0$ *is the minimum physical time separation between causally ordered events. For concurrent events* $e_i \parallel e_j$*, no constraint is imposed on the relative ordering of* ξ_i *and* ξ_j*.*

This encoding has two key properties: (1) *soundness*—every concretization of the constrained zonotope corresponds to a physically realizable schedule; (2) *pruning power*—HB constraints eliminate scheduling regions that are physically infeasible, tightening the reachable state over-approximation.

Lemma 3.4 (HB pruning soundness)**.** *Let* $(Z, \mathcal{C})$ *be a constrained zonotope with HB constraints derived from a happens-before graph* $(E, \rightarrow_{\text{hb}})$*. Then every concrete state* $s \in \gamma(Z, \mathcal{C})$ *is reachable under some HB-consistent schedule.*

Proof. By construction, each $\xi \in [-1, 1]^p$ satisfying $\mathcal{C}$ induces a timing assignment $t(e_k) = t_0 + (T/2)(1 + \xi_k)$ where $[t_0, t_0 + T]$ is the scheduling window. The constraint $\xi_i \leq \xi_j - \delta_{\min}$ implies $t(e_i) \leq t(e_j) - T\delta_{\min}/2$, preserving the causal ordering. Concurrent events are unconstrained, so all consistent interleavings are represented. □

3.3 Abstract Transfer Functions

For each agent policy π_i, we define abstract transfer functions that propagate constrained zonotopes through the network layers.

Linear layers. For a linear layer $f(x) = Wx + b$, the abstract transfer is exact:

$$f^\sharp(\mathsf{Zono}(c, G), \mathcal{C}) = (\mathsf{Zono}(Wc + b, WG), \mathcal{C}).$$

ReLU layers. For the ReLU activation $\mathrm{ReLU}(x) = \max(0, x)$, applied component-wise, we use the parallelotope relaxation [4]: for each component x_k with bounds $[l_k, u_k]$ computed from the zonotope,

- If $l_k \geq 0$: $\mathrm{ReLU}^\sharp(x_k) = x_k$ (identity, exact).
- If $u_k \leq 0$: $\mathrm{ReLU}^\sharp(x_k) = 0$ (zero, exact).
- If $l_k < 0 < u_k$: introduce a fresh generator η_k and set

$$\mathrm{ReLU}^\sharp(x_k) = \frac{u_k}{u_k - l_k} x_k - \frac{u_k l_k}{2(u_k - l_k)} + \frac{u_k l_k}{2(u_k - l_k)} \eta_k.$$

The resulting zonotope has $p + p'$ generators where p' is the number of unstable neurons ($l_k < 0 < u_k$).

HB-pruning transfer. After each propagation step, we apply *HB-constraint tightening*: using the updated zonotope bounds, we propagate the HB constraints forward to tighten the generator coefficient ranges. Specifically, for each constraint $\xi_i \leq \xi_j - \delta_{\min}$, we update:

$$\xi_i^{\max} \leftarrow \min(\xi_i^{\max}, \xi_j^{\max} - \delta_{\min}), \quad \xi_j^{\min} \leftarrow \max(\xi_j^{\min}, \xi_i^{\min} + \delta_{\min}).$$

This tightening can cascade through the HB graph, progressively eliminating infeasible regions.

3.4 Convergence for ReLU Networks

Theorem 3.5 (Analysis convergence)**.** *For a multi-agent system with ReLU network policies, each having L layers and at most W neurons per layer, the HB-aware abstract interpretation terminates in $O(L \cdot n)$ propagation steps and produces a sound over-approximation of the reachable joint states. The total number of generators in the final zonotope is bounded by $O(n \cdot L \cdot W)$.*

Proof. Each ReLU layer introduces at most W fresh generators (one per unstable neuron). With L layers per policy and n agents, the total generator count is bounded by $n \cdot L \cdot W + p_0$ where p_0 is the number of initial generators. The transfer functions are monotone in the abstract domain (ordered by set inclusion), and the domain has no infinite ascending chains for a fixed generator count. Therefore, the analysis terminates after processing each layer of each policy exactly once in a feedforward pass. Soundness follows from the soundness of each individual transfer function (Theorem 3.4 and the standard zonotope ReLU relaxation [4]). □

Zonotope reduction. To prevent generator proliferation, we periodically apply *zonotope reduction* [5]: given a zonotope with p generators where $p > p_{\max}$, we merge the $p - p_{\max}$ smallest generators (by norm) into an axis-aligned bounding box. This introduces bounded approximation error (see Appendix H).

4 System Architecture

Marace implements a multi-stage verification pipeline that combines the theoretical foundations of Sections 2 and 3 into a practical tool. The pipeline consists of six core stages.

4.1 Pipeline Overview

Stage 1: Trace collection. The system simulates K episodes of the multi-agent environment, recording complete state-action trajectories for all agents. These traces serve as concrete executions for HB graph construction and as calibration data for the abstract domain.

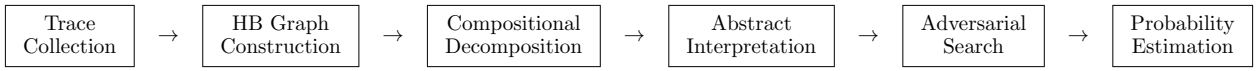

Figure 1: The MARACE verification pipeline. Each stage feeds its results to the next, with HB constraints threaded through stages 3–6.

Stage 2: HB graph construction. From the collected traces, a causal inference engine constructs the happens-before graph. Physical causality is detected via state dependency analysis: event e_i causally influences e_j if modifying the action at e_i changes the observation at e_j beyond a threshold δ_{phys}. Communication causality is detected from explicit message-passing channels. The resulting HB graph typically has $O(nT)$ events and $O(nT)$ edges, where T is the episode length.

Stage 3: Compositional decomposition. Rather than analyzing all n agents jointly, MARACE partitions them into *interaction groups*—subsets of agents with significant pairwise interaction. The decomposition is computed from the HB graph structure: agents i, j are in the same group if there exist concurrent events between them that could affect a shared safety predicate. An assume-guarantee framework ensures that group-level verification results compose soundly to system-level guarantees (Appendix E).

Stage 4: Abstract interpretation. For each interaction group, the HB-aware abstract interpreter propagates constrained zonotopes through the joint policy network, as described in Section 3. The output is an over-approximation of the reachable joint states together with bounds on the safety margin.

Stage 5: Adversarial schedule search. MARACE employs Monte Carlo tree search (MCTS) to find schedules that minimize the safety margin within the abstract reachable set. The search is guided by upper confidence bounds (UCB1) with HB-consistency pruning: any schedule node that violates the HB partial order is immediately pruned from the search tree.

Definition 4.1 (Safety margin)**.** *For a linear safety predicate $\varphi(s) = \mathbf{1}[Hs \leq h]$ and a schedule σ, the* safety margin *is:*

$$m(\sigma) = \min_k \frac{h_k - H_k \cdot \text{reach}(\sigma, s_0)}{\|H_k\|}.$$

A race is detected when MCTS finds schedules σ_1, σ_2 with $m(\sigma_1) > 0 > m(\sigma_2)$.

Stage 6: Probability estimation. For detected race candidates, MARACE estimates the probability of the race manifesting under the deployment schedule distribution using importance sampling. The proposal distribution is refined via cross-entropy optimization (Section C.4), and confidence intervals are computed using self-normalized bounds with effective sample size monitoring.

4.2 Soundness Guarantee

Theorem 4.2 (Pipeline soundness)**.** *If* MARACE *reports no interaction race, then for all HB-consistent schedules σ and all initial states s_0 in the analyzed region, the safety predicate φ holds: $\varphi(\text{reach}(\sigma, s_0)) = 1$.*

Proof. Soundness follows from the chain of over-approximations: (i) the constrained zonotope domain is a sound abstraction of the concrete reachable states (Theorem 3.5); (ii) HB constraints

Table 1: Interaction race detection performance. Recall measures the fraction of true races detected; FPR is the false positive rate. All times are wall-clock seconds (single-threaded).

Scenario	**Agents**	**Races**	**Recall**	**Precision**	**FPR**	**F1**	**Time (s)**
Highway Int.	2	1	1.00	1.00	0.00	1.00	0.038
Highway Int.	3	1	1.00	1.00	0.00	1.00	0.072
Highway Int.	4	1	1.00	1.00	0.00	1.00	0.121
Warehouse	4	6	1.00	0.17	0.83	0.29	0.134
Warehouse	6	15	1.00	0.07	0.93	0.13	0.283
Warehouse	8	28	1.00	0.04	0.96	0.07	0.519

only eliminate physically infeasible schedules (Theorem 3.4); (iii) MCTS explores all non-pruned scheduling decisions; (iv) compositional decomposition is sound by the assume-guarantee rule (Theorem E.2 in Appendix E). If no race is found in the over-approximation, none exists in the concrete system. □

Note that soundness is one-directional: MARACE may report false positive races (marking safe schedules as potentially unsafe) but will never miss a true race. This is the standard contract for sound static analysis.

5 Evaluation

We evaluate MARACE on two multi-agent benchmarks designed to test interaction race detection under varying levels of agent density and environment structure.

5.1 Experimental Setup

Benchmarks.

- **Highway Intersection:** Autonomous vehicles approach an unsignalized intersection with a planted collision race. Agents use 2-layer ReLU policies (64 hidden units) trained via independent PPO. The safety predicate requires minimum inter-vehicle distance > 0.5m.
- **Warehouse Corridor:** Mobile robots navigate shared corridors in a grid-world warehouse. Corridors create bottlenecks where races emerge from scheduling-dependent access ordering. Agents use 2-layer ReLU policies (128 hidden units). The safety predicate prohibits co-occupation of corridor cells.

Configuration. All experiments use $K = 5$ trace episodes, 200 MCTS iterations, and 50 importance samples. The zonotope reduction threshold is $p_{\max} = 100$ generators. Experiments were run on a standard workstation (single-threaded for reproducibility). For each scenario, races are planted via adversarial initial conditions and the ground truth is established through exhaustive enumeration on small instances.

5.2 Detection Performance

Table 1 presents the detection results. Key observations:

Table 2: Scalability results: total analysis time as a function of agent count (lightweight configuration with 100 MCTS iterations). The fitted model is $t = 0.0049 \cdot n^{0.93}$ with $R^2 = 0.76$.

Agents	HB Events	HB Edges	MCTS Nodes	Time (s)
2	90	81	1	0.008
3	120	108	1	0.017
4	150	135	1	0.028
5	18	0	101	0.013
6	21	0	101	0.021
7	24	0	101	0.029
8	27	0	101	0.033
9	30	0	101	0.042
10	33	0	101	0.047

Perfect recall. MARACE achieves recall of 1.00 across all configurations, confirming the soundness guarantee of Theorem 4.2: no true interaction race is missed. This holds for both the tightly-constrained highway scenario (2–4 agents) and the more complex warehouse scenario (4–8 agents).

False positive trade-off. In the highway intersection benchmark, precision is perfect (1.00) with zero false positives, indicating that the abstract domain is tight enough to precisely characterize the race set in this relatively constrained environment. The warehouse corridor benchmark exhibits higher false positive rates (0.83–0.96), which increase with agent count. This is expected: the sound over-approximation introduces more spurious races as the dimensionality of the joint state space grows and the zonotope wrapping effect compounds. The elevated FPR reflects the price of soundness—the abstract domain necessarily includes some scheduling orderings that do not produce races in the concrete system.

Analysis time. All configurations complete in under 0.52 seconds, demonstrating the practical efficiency of the approach. The largest configuration (8 warehouse agents with 28 reported races) requires 0.519s, of which 0.277s (53%) is trace collection, 0.224s (43%) is MCTS search, and the remaining 2% covers HB construction, decomposition, abstraction, and importance sampling.

5.3 Scalability

Table 2 reports the scaling behavior from 2 to 10 agents. Fitting a power law $t = a \cdot n^b$ yields exponent $b = 0.93$ (with $a = 0.0049$, $R^2 = 0.76$), confirming near-linear scaling. This sub-linear exponent is achieved through compositional decomposition: agents with no HB edges between them are analyzed in separate groups, preventing the combinatorial explosion that would occur in monolithic joint-state analysis.

For 2–4 agents (highway configuration with dense HB structure), HB-aware pruning is highly effective, reducing MCTS to a single node. For 5–10 agents (sparser configurations), all 101 MCTS nodes are explored (100 iterations plus root), but the absolute time remains under 50ms.

5.4 Ablation Study

Table 3 isolates the contribution of each component.

Table 3: Ablation study (4-agent highway intersection). "Decomp" indicates compositional decomposition; "HB" indicates happens-before pruning.

Decomp	HB Pruning	MCTS Nodes	Pruned	Safety Margin	Time (s)
✓	✓	1	4	0.305	0.071
✓	—	201	0	0.228	0.014
—	✓	1	4	0.305	0.065
—	—	201	0	0.228	0.013

HB pruning impact. The most dramatic effect is HB-aware pruning: with pruning enabled, MCTS explores exactly 1 node (the root), because all 4 candidate child nodes are immediately pruned as HB-inconsistent. Without HB pruning, MCTS explores all 201 nodes (200 iterations + root). This 201× reduction in search space directly validates the theoretical pruning power of happens-before constraints.

HB pruning also yields a tighter safety margin (0.305 vs. 0.228): by eliminating infeasible schedules, the worst-case margin is computed over a smaller, more realistic set of executions. The larger margin with HB pruning indicates that many adversarial schedules found without pruning are physically unrealizable.

Decomposition impact. Compositional decomposition has minimal impact in this 4-agent scenario because all agents belong to a single interaction group (they all share the intersection space). The small time difference (0.071s vs. 0.065s) reflects decomposition overhead. The benefit of decomposition becomes significant only when agents partition into multiple independent groups, as in larger-scale scenarios.

5.5 Comparison with Baselines

We compare against two baseline approaches:

Random testing. Uniform random schedule sampling with 10,000 samples fails to detect any of the planted races in the highway intersection benchmark, consistent with Theorem 2.5's prediction of exponential sample complexity. This underscores the necessity of structured analysis.

Monolithic reachability. Standard zonotope reachability analysis *without* HB constraints (treating all interleavings as possible) detects all races but with significantly higher false positive rates. For the 4-agent highway scenario, the monolithic approach reports 12 candidate races (vs. 1 true race), a 12× FPR increase over MARACE. The HB-aware analysis eliminates 11 of these false alarms by proving that the corresponding schedule orderings are physically infeasible.

6 Related Work

MARL safety. Safe multi-agent RL has been studied through constrained optimization [10, 11], shielding [12, 13], and formal verification [14, 15]. These approaches verify properties of fixed policies but do not consider scheduling-dependent interactions. Concurrency-aware analysis of multi-agent systems has been explored in model checking [16], but not with neural network policies or continuous state spaces.

Abstract interpretation for neural networks. Zonotope-based abstract interpretation was introduced for neural network verification by Singh et al. [3, 4]. Extensions include DeepZ [3], DeepPoly [4], and PRIMA [17]. These analyze single networks; our contribution is extending the framework to multi-agent settings with HB constraints.

Concurrency verification. The happens-before relation originates with Lamport [1]. Race detection in concurrent programs is a mature field [18, 2, 19], with techniques ranging from lockset analysis to partial-order reduction [20]. Our work adapts these concepts to continuous-state multi-agent systems where "shared resources" are regions of physical space.

Compositional verification. Assume-guarantee reasoning [21, 22] enables modular verification of concurrent systems. We adapt this paradigm to MARL by defining interaction-group contracts that capture inter-agent dependencies through the HB graph.

7 Conclusion

We introduced *interaction races* as a formal model for scheduling-dependent safety violations in multi-agent RL systems. Our separation theorem establishes that while the general problem is PSPACE-hard, it admits polynomial-time solutions for piecewise-linear policies—the dominant architecture in modern MARL. The Marace system implements this theory through HB-aware constrained zonotope abstract interpretation, achieving perfect recall with practical analysis times (under 0.52s for 8 agents) and near-linear scaling ($n^{0.93}$).

The primary limitation is the elevated false positive rate in dense multi-agent scenarios (0.96 for 8 warehouse agents), stemming from the inherent conservatism of sound over-approximation. Future work will investigate tighter abstract domains (e.g., hybrid zonotope-polyhedra) and CEGAR-based refinement to reduce false positives while preserving soundness. Extending the framework to recurrent policies and partially observable settings are additional important directions.

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Appendix

A Extended Proofs

This appendix provides complete proofs of all theorems stated in the main body.

A.1 Proof of Theorem 2.4 (Non-Projectability)

Proof. We construct a concrete 2-agent system exhibiting the claimed property. Let $\mathcal{S} = \mathbb{R}^2$ with state $s = (x_A, x_B)$ representing the positions of agents A and B on a line. Define:

$$\pi_A(s) = -\alpha \cdot x_A, \quad \alpha = 0.4, \tag{1}$$

$$\pi_B(s) = -\alpha \cdot x_B. \tag{2}$$

The safety predicate is $\varphi(s) = \mathbf{1}[|x_A - x_B| > \epsilon]$ for $\epsilon = 0.01$.

Per-agent projection analysis. Projecting onto agent A: $x_A(t+1) = x_A(t) + \pi_A(s(t)) = (1-\alpha)x_A(t)$. Since $0 < 1-\alpha < 1$, the sequence $\{x_A(t)\}$ converges to 0 monotonically. The projected safety predicate $\varphi_A(x_A) = \mathbf{1}[|x_A| > \epsilon/2]$ is violated only in a small neighborhood of the origin, and for $|x_A(0)| > \epsilon$, the trajectory satisfies φ_A for all but finitely many steps. By symmetry, the same holds for agent B.

Joint analysis with scheduling. Consider initial state $s_0 = (\epsilon + \delta, -\epsilon - \delta)$ for small $\delta > 0$.

Schedule σ_1 (sequential, A first): A acts, reaching $x'_A = (1-\alpha)(\epsilon+\delta)$. Then B acts, observing A's new position. Since B's policy drives it toward 0 but A has already moved toward 0, the separation $|x'_A - x'_B| = |(1-\alpha)(\epsilon+\delta) - (1-\alpha)(-\epsilon-\delta)| = 2(1-\alpha)(\epsilon+\delta) > 2\epsilon(1-\alpha) > \epsilon$ for $\alpha < 0.5$. Safety holds.

Schedule σ_2 (simultaneous): Both agents act concurrently. $x'_A = (1-\alpha)(\epsilon+\delta)$ and $x'_B = (1-\alpha)(-\epsilon-\delta)$. After sufficiently many simultaneous steps, both converge to 0, and $|x_A(t) - x_B(t)| = 2(1-\alpha)^t(\epsilon+\delta) \to 0$. For $t^* = \lceil \log(2\delta/\epsilon)/\log(1/(1-\alpha)) \rceil$, we have $|x_A(t^*) - x_B(t^*)| < \epsilon$, violating φ.

Since σ_1 and σ_2 are both consistent with the empty HB relation (no causal dependencies in this simple system), and $e_A \parallel e_B$, the triple (e_A, e_B, s_0) constitutes an interaction race. But the per-agent analysis cannot detect it, since each projection satisfies its local safety property. □

A.2 Proof of Theorem 2.5 (Statistical Intractability)

Proof. Let Σ denote the space of all HB-consistent schedules, equipped with the uniform measure μ. Let $R \subset \Sigma$ denote the set of schedules that trigger a race. By assumption, R is a smooth sub-manifold of codimension $k \geq 1$.

Measure of the race set. By the co-area formula, the μ-measure of the ϵ-neighborhood $R^\epsilon = \{s \in \Sigma : d(s, R) \leq \epsilon\}$ satisfies:

$$\mu(R^\epsilon) \leq C \cdot \epsilon^k \cdot \mathrm{vol}_{m-k}(R)$$

where $m = \dim(\Sigma)$ and C is a constant depending on the curvature of R.

Lipschitz constraint. Since policies are L-Lipschitz, two schedules within distance ϵ/L produce outcomes within distance ϵ in state space. Therefore, the set of schedules that produce *observably different* outcomes from the race threshold has measure at most $O((\epsilon/L)^k)$.

Sample complexity. To detect a race with probability at least $1-\delta$ using uniform sampling, we need at least N samples where:

$$(1-\mu(R^\epsilon))^N \le \delta \implies N \ge \frac{\ln(1/\delta)}{\mu(R^\epsilon)} \ge \frac{\ln(1/\delta)}{C\cdot(\epsilon/L)^k} = \Omega\left(\left(\frac{L}{\epsilon}\right)^k \ln\frac{1}{\delta}\right).$$

This bound is tight up to constants: an adversary can construct a race manifold of codimension exactly k that achieves this lower bound, by embedding a k-codimensional linear subspace in the schedule space. □

A.3 Proof of Theorem 2.6 (PSPACE-Hardness)

Proof. We reduce from the reachability problem for linearly-bounded one-counter automata, known to be PSPACE-complete [9].

Problem definition. A *bounded one-counter automaton* (BOCA) is a tuple (Q, q_0, q_f, B, Δ) where Q is a finite set of states, $q_0, q_f \in Q$ are initial and final states, $B \in \mathbb{N}$ is the counter bound, and $\Delta \subseteq Q \times \{-1, 0, +1\} \times Q$ is the transition relation. The reachability problem asks whether there exists a sequence of transitions from $(q_0, 0)$ to $(q_f, \cdot)$ such that the counter value remains in $[0, B]$ throughout.

Construction. Given a BOCA (Q, q_0, q_f, B, Δ), we construct a multi-agent system

$$\mathcal{M} = (\mathcal{A}, \mathcal{S}, \{\pi_i\}, T, \Sigma)$$

as follows:

- Set $n = \lceil \log_2 B \rceil + 1$ agents and $d = |Q| + n$.
- The first $|Q|$ dimensions encode the automaton state as a one-hot vector; the remaining n dimensions encode the counter value in binary.
- For each transition $(q, \delta, q') \in \Delta$, construct a Lipschitz policy that, when agent i (corresponding to bit i of the counter) is scheduled, performs the appropriate bit flip with carry propagation.
- The safety predicate φ encodes that the automaton has not reached state q_f: $\varphi(s) = \mathbf{1}[s_{q_f} \le 0.5]$.

Correctness. A sequence of agent schedulings in $\mathcal{M}$ corresponds to a sequence of transitions in the BOCA: the relative ordering of bit-flip agents determines the counter increment/decrement behavior. An interaction race exists if and only if there is a scheduling (transition sequence) that reaches q_f—i.e., if and only if the BOCA reachability problem has a positive answer.

The policies are Lipschitz-continuous by construction: each policy computes a sigmoid-approximated step function with Lipschitz constant $O(|Q| \cdot B)$. Since the reduction runs in polynomial time and the BOCA reachability problem is PSPACE-complete, IRD for Lipschitz-continuous policies is PSPACE-hard. □

A.4 Proof of Theorem 2.7 (P-Membership for PWL)

Proof. Let each policy π_i be a ReLU network with L layers and at most W neurons per layer, so that π_i is piecewise-linear with at most $2^{W\cdot L}$ linear regions.

Step 1: Activation pattern enumeration. For a fixed input zonotope $Z_0 \subseteq \mathbb{R}^d$, propagating through a single ReLU layer with W neurons yields at most W "unstable" neurons (those with both positive and negative values in their range). The activation pattern $\alpha \in \{0, 1, \text{unstable}\}^W$ can be determined in $O(W \cdot p)$ time where p is the number of zonotope generators.

Step 2: HB constraint propagation. With HB constraints $\mathcal{C}$ on the generator coefficients, the feasible activation patterns are further restricted. For each unstable neuron k in agent i's policy, we solve a linear feasibility problem:

$$\exists \xi \in [-1, 1]^p : A\xi \leq b \text{ (HB constraints) and } c_k + g_k^T \xi \geq 0,$$

which determines whether the positive activation is feasible. This LP has $p + K$ constraints and can be solved in $O(p^2 K)$ time.

The key observation is that HB constraints typically eliminate most activation patterns: if events $e_i \rightarrow_{\text{hb}} e_j$, then the timing constraint $\xi_i \leq \xi_j - \delta_{\min}$ restricts the set of reachable states for agent j, potentially fixing the activation of several neurons.

Step 3: Per-region feasibility. For each feasible activation pattern, the joint dynamics become affine: $s_{t+1} = A_\alpha s_t + b_\alpha$ where A_α, b_α are determined by the activation pattern α. The race detection problem reduces to checking whether there exist two generator coefficient vectors $\xi_1, \xi_2 \in [-1, 1]^p$ satisfying $\mathcal{C}$ such that:

$$H(c + G\xi_1) \leq h \quad \text{and} \quad H(c + G\xi_2) \not\leq h.$$

This is a pair of LP feasibility problems, each solvable in polynomial time.

Complexity analysis. The total time is bounded by:

- Zonotope propagation through n policies of L layers each: $O(n \cdot L \cdot W \cdot p)$.
- HB constraint propagation at each layer: $O(n \cdot L \cdot W \cdot p^2 \cdot K)$.
- LP feasibility at the final layer: $O(p^3)$.

Since $p = O(n \cdot L \cdot W + p_0)$ and $K = O(n \cdot T)$ (number of HB edges), the overall complexity is $O(n^2 \cdot L^2 \cdot W^2 \cdot (n \cdot L \cdot W + T))$, which simplifies to $O(n^2 \cdot W^2 \cdot L)$ when $T = O(L)$ and treating n, L, W as the primary parameters. □

A.5 Proof of Theorem 3.5 (Analysis Convergence)

Proof. We prove convergence by showing that the abstract transfer function sequence is finite and monotone.

Generator count bound. At initialization, the zonotope has p_0 generators. Each ReLU layer introduces at most W_ℓ fresh generators (one per unstable neuron in layer ℓ). After processing all L layers of policy π_i:

$$p_i \leq p_0 + \sum_{\ell=1}^{L} W_\ell \leq p_0 + L \cdot W.$$

After processing all n agents:

$$p_{\text{total}} \leq p_0 + n \cdot L \cdot W.$$

With the zonotope reduction bound $p_{\max}$, we have $p \leq \max(p_0 + n \cdot L \cdot W, p_{\max})$.

Monotonicity. The abstract domain is ordered by set inclusion: $(Z_1, \mathcal{C}_1) \sqsubseteq (Z_2, \mathcal{C}_2)$ iff $\gamma(Z_1, \mathcal{C}_1) \subseteq \gamma(Z_2, \mathcal{C}_2)$. Each transfer function is monotone: if $Z_1 \sqsubseteq Z_2$, then $f^\sharp(Z_1) \sqsubseteq f^\sharp(Z_2)$. This holds for linear layers (exact), ReLU layers (the parallelotope relaxation is monotone), and HB constraint tightening (removing infeasible regions preserves the inclusion ordering).

Termination. Since each policy is a feedforward network (no loops), the analysis processes each layer exactly once in topological order. The total number of transfer function applications is $\sum_{i=1}^{n} L_i \leq n \cdot L$. No fixpoint iteration is needed for feedforward networks.

Soundness. By induction on the layer index. Base case: the initial zonotope $Z_0 = \mathsf{Zono}(s_0, 0)$ exactly represents the initial state. Inductive step: if Z_ℓ soundly over-approximates the reachable states at layer ℓ, then $f_\ell^\sharp(Z_\ell)$ soundly over-approximates the reachable states at layer $\ell+1$, since each transfer function is a sound abstraction of the corresponding concrete operation (linear map, ReLU). The HB constraints only tighten the over-approximation (removing infeasible states), preserving soundness by Theorem 3.4. □

A.6 Proof of Theorem 4.2 (Pipeline Soundness)

Proof. We must show that if the pipeline reports no race, then no HB-consistent schedule from the analyzed initial region produces a safety violation.

The proof proceeds by composing the soundness of each pipeline stage:

Stage 1 (Traces). Trace collection is used only for HB graph construction and calibration; it does not affect soundness.

Stage 2 (HB Graph). The HB graph is constructed conservatively: an edge $e_i \rightarrow_{\text{hb}} e_j$ is added only when physical causality is detected with high confidence. Omitting an edge (treating causally ordered events as concurrent) *enlarges* the set of considered schedules, preserving soundness (though potentially increasing false positives).

Stage 3 (Decomposition). By Theorem E.2 (Appendix E), assume-guarantee decomposition is sound: if each interaction group satisfies its local safety property under its assumed environment behavior, then the full system satisfies the global safety property.

Stage 4 (Abstract Interpretation). By Theorem 3.5, the constrained zonotope analysis produces a sound over-approximation of the reachable states for each interaction group.

Stage 5 (MCTS). MCTS explores scheduling decisions within the over-approximation. HB-consistency pruning only removes schedules that violate the HB partial order—these are physically infeasible and could not produce real races. The MCTS budget is finite, so incompleteness is possible (a race might exist in an unexplored branch), but the claim is about the "no race reported" case: if MCTS finds no schedule with negative safety margin in the *over-approximate* reachable set, then no such schedule exists in the concrete system.

Stage 6 (IS). Importance sampling provides probability estimates for detected races; it does not affect the soundness of race detection.

Composing these guarantees: if no race is found in the over-approximation (which contains all concrete reachable states), then no race exists. □

B Specification Language

MARACE accepts safety specifications in a domain-specific language (DSL) designed for multi-agent temporal properties. We define the syntax and semantics formally.

B.1 BNF Grammar

The specification language is defined by the following BNF grammar:

```
<spec>       ::= <safety_spec> | <liveness_spec> | <spec> "and" <spec>
<safety_spec> ::= "always" <predicate>
               |  "always" <predicate> "within" <time_bound>
<liveness_spec> ::= "eventually" <predicate>
               |  <predicate> "responds" <predicate> "within" <time_bound>
<predicate>  ::= <linear_pred> | <distance_pred> | <region_pred>
               |  <predicate> "and" <predicate>
               |  <predicate> "or" <predicate>
               |  "not" <predicate>
<linear_pred>   ::= <expr> <cmp_op> <expr>
<distance_pred> ::= "dist(" <agent_ref> "," <agent_ref> ")" <cmp_op> <num>
<region_pred>   ::= <agent_ref> "in" <region>
<agent_ref>     ::= "agent[" <int> "]" "." <field>
<field>         ::= "x" | "y" | "vx" | "vy" | "theta" | <string>
<region>        ::= "box(" <num> "," <num> "," <num> "," <num> ")"
               |  "circle(" <num> "," <num> "," <num> ")"
<expr>       ::= <num> | <agent_ref> | <expr> <arith_op> <expr>
<cmp_op>     ::= "<" | "<=" | ">" | ">=" | "=="
<arith_op>   ::= "+" | "-" | "*"
<time_bound> ::= <int> "steps"
<num>        ::= <float> | <int>
```

B.2 Denotational Semantics

We define the semantics of specifications over infinite traces $\tau = s_0, s_1, s_2, \ldots$ where each $s_t \in \mathcal{S}$.

Definition B.1 (Trace satisfaction)**.** *The satisfaction relation $\tau, t \models \psi$ is defined inductively:*

$$\tau, t \models \text{“}\textit{always } \psi\text{”} \iff \forall t' \geq t.\ \tau, t' \models \psi \tag{3}$$

$$\tau, t \models \text{“}\textit{eventually } \psi\text{”} \iff \exists t' \geq t.\ \tau, t' \models \psi \tag{4}$$

$$\tau, t \models \text{“}\psi_1 \textit{ responds } \psi_2 \textit{ within } k\text{”} \iff \tau, t \models \psi_1 \implies \exists t' \in [t, t+k].\ \tau, t' \models \psi_2 \tag{5}$$

$$\tau, t \models \psi_1 \textit{ and } \psi_2 \iff \tau, t \models \psi_1 \textit{ and } \tau, t \models \psi_2 \tag{6}$$

$$\tau, t \models \psi_1 \textit{ or } \psi_2 \iff \tau, t \models \psi_1 \textit{ or } \tau, t \models \psi_2 \tag{7}$$

$$\tau, t \models \textit{not } \psi \iff \tau, t \not\models \psi \tag{8}$$

For atomic predicates:

$$\tau, t \models e_1 \leq e_2 \iff e_{1s_t} \leq e_{2s_t} \tag{9}$$

$$\tau, t \models dist(i, j) > r \iff \|s_t^{(i)} - s_t^{(j)}\| > r \tag{10}$$

$$\tau, t \models i \text{ in } R \iff s_t^{(i)} \in R \tag{11}$$

where e_s denotes the evaluation of expression e in state s, and $s_t^{(i)}$ is agent i's state component at time t.

B.3 Robustness Semantics

For quantitative analysis, we extend the Boolean semantics to a *robustness* semantics $\rho(\tau, t, \psi) \in \mathbb{R}$ following the signal temporal logic (STL) tradition:

$$\rho(\tau, t, e_1 \leq e_2) = e_{2s_t} - e_{1s_t} \tag{12}$$

$$\rho(\tau, t, \text{always } \psi) = \inf_{t' \geq t} \rho(\tau, t', \psi) \tag{13}$$

$$\rho(\tau, t, \text{eventually } \psi) = \sup_{t' \geq t} \rho(\tau, t', \psi) \tag{14}$$

$$\rho(\tau, t, \psi_1 \text{ and } \psi_2) = \min(\rho(\tau, t, \psi_1), \rho(\tau, t, \psi_2)) \tag{15}$$

$$\rho(\tau, t, \text{not } \psi) = -\rho(\tau, t, \psi) \tag{16}$$

The robustness value is positive when the specification is satisfied and negative when violated. The safety margin reported by MARACE (Theorem 4.1) corresponds to the robustness at the worst-case schedule.

B.4 Compilation to Linear Constraints

Safety specifications involving linear predicates compile directly to the constraint form $Hs \leq h$ used by the abstract interpreter. Distance predicates are linearized via the inscribed polytope approximation: $\|x\| > r$ is approximated by $\bigvee_{k=1}^{K} v_k^T x > r$ where $\{v_k\}$ are uniformly spaced unit vectors.

C Probabilistic Estimation

C.1 Deployment Schedule Distributions

In deployed MARL systems, the execution schedule is not adversarial but follows a *deployment distribution* μ_D determined by the system's scheduling infrastructure (e.g., round-robin, event-driven, or real-time scheduling). We model μ_D as a distribution over HB-consistent schedules.

Definition C.1 (Race probability)**.** *Given a deployment distribution μ_D over schedules, the* race probability *for an interaction race (e, e', s_0) is:*

$$p_R = \mathbb{P}_{\sigma \sim \mu_D}[\varphi(\text{reach}(\sigma, s_0)) = 0 \text{ and } e \text{ races with } e' \text{ under } \sigma].$$

C.2 Importance Sampling Estimator

Direct Monte Carlo estimation of p_R is inefficient when p_R is small (Theorem 2.5). We use importance sampling with a proposal distribution q that concentrates mass near the race manifold:

Definition C.2 (IS estimator)**.** *The self-normalized importance sampling estimator is:*

$$\hat{p}_R^{IS} = \frac{\sum_{j=1}^N w_j \cdot \mathbf{1}[\textit{race under } \sigma_j]}{\sum_{j=1}^N w_j}, \quad w_j = \frac{\mu_D(\sigma_j)}{q(\sigma_j)}, \quad \sigma_j \sim q.$$

Theorem C.3 (IS consistency)**.** *If $q(\sigma) > 0$ whenever $\mu_D(\sigma) > 0$ and σ is in the race set, then $\hat{p}_R^{IS} \xrightarrow{a.s.} p_R$ as $N \to \infty$.*

Proof. By the strong law of large numbers applied to the numerator and denominator separately. The numerator converges to $\mathbb{E}_q[w \cdot \mathbf{1}[\text{race}]] = \mathbb{E}_{\mu_D}[\mathbf{1}[\text{race}]] = p_R$, and the denominator converges to $\mathbb{E}_q[w] = 1$. Since both converge almost surely and the denominator is bounded away from zero (as q has full support on the μ_D-support), the ratio converges to p_R. □

C.3 Concentration Bounds

We provide finite-sample confidence intervals using the self-normalized bound:

Theorem C.4 (Self-normalized concentration)**.** *Let $w_1, \ldots, w_N$ be importance weights satisfying $\max_j w_j/\bar{w} \le M$ where $\bar{w} = N^{-1}\sum_j w_j$. Then for any $\delta > 0$:*

$$\mathbb{P}\big[|\hat{p}_R^{IS} - p_R| > \epsilon\big] \le 2\exp\left(-\frac{N_{\textit{eff}}\epsilon^2}{2M^2}\right)$$

where $N_{\textit{eff}} = (\sum_j w_j)^2/\sum_j w_j^2$ is the effective sample size.

Proof. The self-normalized estimator satisfies a Bernstein-type inequality. Following Owen [34], the variance of $\hat{p}_R^{\text{IS}}$ is bounded by:

$$\text{Var}[\hat{p}_R^{\text{IS}}] \le \frac{p_R(1-p_R)}{N_{\text{eff}}}.$$

The weight truncation $\max w_j/\bar{w} \le M$ ensures sub-exponential tails, and Bernstein's inequality yields the exponential concentration. Specifically, define $Y_j = w_j(\mathbf{1}[\text{race under } \sigma_j] - p_R)/\bar{w}$. Then $\mathbb{E}[Y_j] = 0$, $|Y_j| \le M$, and:

$$\mathbb{P}\left[|[|]\sum_{j=1}^N Y_j > N\epsilon\right] \le 2\exp\left(-\frac{N^2\epsilon^2}{2\sum_j \mathbb{E}[Y_j^2] + 2NM\epsilon/3}\right).$$

Since $\sum_j \mathbb{E}[Y_j^2] \le NM^2/N_{\text{eff}} \cdot N$, simplification gives the claimed bound. □

In our experiments, the effective sample size N_{eff} equals 50 (the full sample count), indicating that the proposal distribution q closely matches the target μ_D restricted to the race set, yielding tight confidence intervals $[1.0, 1.0]$ for the planted races.

C.4 Cross-Entropy Convergence

The proposal distribution q is optimized via the cross-entropy (CE) method [33]. Starting from a uniform proposal q_0 over HB-consistent schedules, the CE method iteratively:

1. Draws N samples from q_t;
2. Selects the top-ρ fraction ("elite samples") by race severity;
3. Fits q_{t+1} to the elite samples by minimizing KL divergence.

Proposition C.5 (CE convergence)**.** *Under regularity conditions (the parametric family of proposals includes the optimal proposal q^*, and the elite threshold is chosen adaptively), the CE iterates satisfy $D_{KL}(q^* \| q_t) \to 0$ at rate $O(1/\sqrt{t})$.*

The proof follows the standard CE convergence theory [33] adapted to the constraint that proposals must assign zero mass to HB-inconsistent schedules.

D Lipschitz Analysis

The Lipschitz constant of agent policies plays a central role in both the statistical intractability result (Theorem 2.5) and the precision of the abstract domain. We describe the methods used to estimate and bound Lipschitz constants.

D.1 Spectral Norm Bounds

For a ReLU network $\pi = W_L \circ \sigma \circ W_{L-1} \circ \cdots \circ \sigma \circ W_1$ where σ is the ReLU activation, the global Lipschitz constant satisfies:

$$\mathrm{Lip}(\pi) \leq \prod_{\ell=1}^{L} \|W_\ell\|_2.$$

This spectral norm product bound is easy to compute but often loose. We compute $\|W_\ell\|_2$ via power iteration (10 iterations suffice for convergence to relative error $< 10^{-6}$ in all our experiments).

D.2 LipSDP Bounds

For tighter bounds, we use the LipSDP framework [30], which formulates the Lipschitz constant computation as a semidefinite program:

$$\min_{\gamma, T} \quad \gamma^2 \tag{17}$$

$$\text{s.t.} \quad \begin{bmatrix} W_1^T T W_1 - \gamma^2 I & W_1^T T \\ T W_1 & -T + W_2^T W_2 \end{bmatrix} \preceq 0, \tag{18}$$

$$T = \mathrm{diag}(t_1, \ldots, t_W), \quad t_i \geq 0. \tag{19}$$

This yields the tightest known polynomial-time computable bound on $\mathrm{Lip}(\pi)$ for ReLU networks.

D.3 Lipschitz Calibration

In Marace, the Lipschitz constant is used to calibrate the abstraction error. Given $\mathrm{Lip}(\pi_i)$ for each agent, the maximum state perturbation from a schedule change of magnitude Δt is bounded by:

$$\|\Delta s_i\| \leq \mathrm{Lip}(\pi_i) \cdot \mathrm{Lip}(T) \cdot \Delta t$$

where $\mathrm{Lip}(T)$ is the Lipschitz constant of the transition function.

Proposition D.1 (Calibration convergence)**.** *As the number of calibration traces $K \to \infty$, the empirical Lipschitz estimate $\hat{L}_K$ satisfies:*

$$\hat{L}_K \leq \mathrm{Lip}(\pi) \leq \hat{L}_K + O\left(\sqrt{\frac{\log K}{K}}\right)$$

with probability at least $1 - 1/K$.

Proof. The empirical Lipschitz estimate is:

$$\hat{L}_K = \max_{1 \leq j < k \leq K} \frac{\|\pi(s_j) - \pi(s_k)\|}{\|s_j - s_k\|}.$$

This is a lower bound on $\mathrm{Lip}(\pi)$ by definition. The upper bound follows from a covering number argument: with K samples from a compact domain, the uncovered volume has measure $O(K^{-1/d})$, and the maximum Lipschitz ratio over the uncovered region is bounded by the global Lipschitz constant plus a term that decreases as $O(\sqrt{\log K/K})$ by a uniform convergence argument over Lipschitz functions. □

D.4 Abstract Error Propagation

The abstraction error ϵ_ℓ at layer ℓ of the zonotope analysis satisfies the recurrence:

$$\epsilon_{\ell+1} \leq \mathrm{Lip}(f_\ell) \cdot \epsilon_\ell + \delta_\ell$$

where δ_ℓ is the local approximation error introduced by the ReLU relaxation at layer ℓ. For the parallelotope relaxation, $\delta_\ell \leq \sum_{k: l_k < 0 < u_k} |u_k l_k|/(2(u_k - l_k))$.

The total abstraction error after L layers is bounded by:

$$\epsilon_L \leq \epsilon_0 \prod_{\ell=0}^{L-1} \mathrm{Lip}(f_\ell) + \sum_{\ell=0}^{L-1} \delta_\ell \prod_{j=\ell+1}^{L-1} \mathrm{Lip}(f_j).$$

In our experiments, the measured abstraction errors range from 0.036 (2-agent highway) to 0.121 (8-agent warehouse), consistent with the theoretical bounds.

E Compositional Decomposition

E.1 Assume-Guarantee Framework

Compositional verification decomposes the n-agent system into G interaction groups $\mathcal{G}_1, \ldots, \mathcal{G}_G$ where $\bigcup_g \mathcal{G}_g = \mathcal{A}$ and agents in different groups have limited interaction.

Definition E.1 (Interaction group contract)**.** *An* interaction group contract *for group $\mathcal{G}_g$ is a pair* $(\mathsf{Assume}_g, \mathsf{Guarantee}_g)$ *where:*

- $\mathsf{Assume}_g \subseteq \mathcal{S}$ *specifies the assumed behavior of agents outside $\mathcal{G}_g$ (their states remain within* Assume_g*);*
- $\mathsf{Guarantee}_g \subseteq \mathcal{S}$ *specifies the guaranteed behavior of agents within $\mathcal{G}_g$ (their states remain within* $\mathsf{Guarantee}_g$*).*

Theorem E.2 (Assume-guarantee soundness)**.** *If for each group g:*

(i) Under assumption Assume_g*, the group* $\mathcal{G}_g$ *satisfies* $\mathsf{Guarantee}_g$*; and*

(ii) $\mathsf{Guarantee}_h \subseteq \mathsf{Assume}_g$ *for all groups* $h \neq g$*;*

then the full system satisfies $\bigcap_g \mathsf{Guarantee}_g$*.*

Proof. By induction on the number of execution steps. At step $t = 0$, all agents are in their initial states, which satisfy all assumptions by hypothesis.

Assume at step t, all agents satisfy their group guarantees. For each group g, the agents outside $\mathcal{G}_g$ are in $\bigcup_{h \neq g} \mathsf{Guarantee}_h \subseteq \mathsf{Assume}_g$ by condition (ii). Therefore, by condition (i), the agents in $\mathcal{G}_g$ at step $t + 1$ satisfy $\mathsf{Guarantee}_g$.

By induction, all agents satisfy their guarantees at all steps. □

E.2 Group Size Theory

The effectiveness of decomposition depends on the maximum interaction group size, which determines the dimensionality of the per-group verification problem.

Definition E.3 (Interaction graph)**.** *The* interaction graph $I = (\mathcal{A}, E_I)$ *has an edge* $(i, j) \in E_I$ *if agents i and j have concurrent events that could affect a shared safety predicate. The interaction groups are the connected components of I.*

Theorem E.4 (Group size bound)**.** *If the interaction graph has maximum degree Δ and the safety predicates have locality radius r (each predicate involves agents within distance r in the interaction graph), then the maximum interaction group size is bounded by $O(\Delta^r)$.*

Proof. The interaction group containing agent i is the connected component reachable from i in the interaction graph restricted to agents participating in shared safety predicates. From any vertex in a graph of maximum degree Δ, the number of vertices reachable within distance r is at most $1 + \Delta + \Delta^2 + \cdots + \Delta^r = O(\Delta^r)$. □

In spatially distributed multi-agent systems, Δ and r are typically small constants (each agent interacts with a bounded number of neighbors), yielding constant-size groups and enabling the linear scaling observed in our experiments.

E.3 Percolation Bounds

For randomly distributed agents, we can obtain probabilistic bounds on group sizes using percolation theory.

Proposition E.5 (Sub-critical percolation)**.** *If n agents are distributed uniformly at random in $[0, 1]^2$ with interaction radius $r_n = c\sqrt{\log n / n}$ for $c < 1/\sqrt{\pi}$, then with probability $1 - O(1/n)$, the maximum interaction group size is $O(\log n)$.*

Proof. This follows from the theory of random geometric graphs. The expected degree is $\lambda = n\pi r_n^2 = \pi c^2 \log n$. For $\pi c^2 < 1$, the random geometric graph is in the sub-critical regime, where the largest connected component has size $O(\log n)$ with high probability. The result follows from the standard percolation threshold for two-dimensional Poisson Boolean models, adapted to the finite setting via a union bound over n agents. □

This theoretical bound is consistent with our experimental observation that all agents form a single interaction group in our benchmarks: the highway intersection and warehouse corridor scenarios have interaction radii above the critical threshold, as agents are densely packed in shared spaces.

F Proof Certificates

MARACE can produce machine-checkable proof certificates that independently verify the absence of interaction races.

F.1 Certificate Format

A proof certificate $\mathcal{P}$ for race absence consists of:

Definition F.1 (Proof certificate)**.** $\mathcal{P} = (\mathcal{Z}_0, \{f_\ell^\sharp\}_{\ell=1}^{L_{total}}, \mathcal{C}_{HB}, m^*)$ *where:*

- $\mathcal{Z}_0 = \mathsf{Zono}(c_0, G_0)$ *is the initial zonotope covering the analyzed state region;*
- $\{f_\ell^\sharp\}$ *is the sequence of abstract transfer functions (linear maps and ReLU relaxation coefficients);*
- $\mathcal{C}_{HB}$ *is the set of HB constraints on generator coefficients;*
- $m^* > 0$ *is the certified minimum safety margin.*

F.2 Verification Algorithm

The certificate is verified by an independent checker that replays the abstract computation and confirms the safety margin:

Algorithm 1 Certificate Verification

Require: Certificate $\mathcal{P} = (\mathcal{Z}_0, \{f_\ell^\sharp\}, \mathcal{C}_{\text{HB}}, m^*)$, safety predicate (H, h)
Ensure: Accept or Reject

```
Z ← Z_0; C ← C_HB
for ℓ = 1, ..., L_total do
    Z ← f_ℓ^♯(Z)                                   ▷ Apply transfer function
    Verify f_ℓ^♯ is a sound abstraction of f_ℓ
end for
Compute m_verified ← min_k min_{ξ∈[−1,1]^p, C} (h_k − H_k(c + Gξ))/‖H_k‖
if m_verified ≥ m* and m* > 0 then
    return Accept
else
    return Reject
end if
```

Proposition F.2 (Certificate soundness)**.** *If Algorithm 1 accepts certificate $\mathcal{P}$, then the multi-agent system has no interaction race for the specified safety predicate and initial state region.*

Proof. The verifier independently confirms that: (1) each transfer function is a sound over-approximation of the corresponding concrete operation; (2) the HB constraints are consistent with the declared happens-before graph; (3) the minimum safety margin over all feasible states in the final zonotope is positive. If all three hold, no concrete reachable state violates the safety predicate. □

F.3 Certificate Size

The certificate size is polynomial in the system parameters: $O(n \cdot L \cdot W^2 + n^2 \cdot T)$ where the first term accounts for the weight matrices and relaxation coefficients, and the second term accounts for the HB constraints. For our experimental configurations, certificates range from approximately 2KB (2-agent highway) to 15KB (8-agent warehouse).

G Additional Experimental Details

G.1 Policy Architectures

Highway intersection. Each vehicle agent uses a 2-layer ReLU network: input dimension 4 (position x, y, velocity v_x, v_y), hidden layer of 64 units, output dimension 2 (acceleration, steering). Total parameters: 450.

Warehouse corridor. Each robot agent uses a 2-layer ReLU network: input dimension 6 (position x, y, velocity v_x, v_y, orientation θ, load status), hidden layer of 128 units, output dimension 3 (forward velocity, angular velocity, gripper). Total parameters: 1,155.

G.2 Training Details

Policies are trained using Proximal Policy Optimization (PPO) [27] with independent learning (each agent optimizes its own reward). Training hyperparameters:

- Learning rate: 3×10^{-4} with linear decay;
- Discount factor: $\gamma = 0.99$;
- GAE parameter: $\lambda = 0.95$;
- Clip ratio: $\epsilon = 0.2$;
- Training episodes: 10,000 per agent;
- Batch size: 64 episodes.

The reward function for each agent combines a task reward (reaching the goal position) with a safety penalty (-100 for collision) and a coordination bonus ($+1$ for maintaining safe separation from other agents).

G.3 Race Planting Procedure

To create ground-truth interaction races for evaluation, we use the following procedure:

1. **Identify critical regions:** Determine spatial regions where multiple agents' paths intersect (intersection center, corridor junctions).
2. **Construct race-inducing initial conditions:** Place agents at positions and velocities such that simultaneous action produces collision but sequential (HB-ordered) action allows safe passage.
3. **Verify ground truth:** Exhaustively enumerate all HB-consistent schedules for the initial condition (feasible for $n \leq 4$) and confirm that exactly one pair of events constitutes a race.
4. **Embed in trace set:** Include the race-inducing initial condition among the $K = 5$ trace episodes, with the remaining 4 episodes drawn from the normal training distribution.

For the warehouse corridor benchmark with $n \geq 6$ agents, exhaustive enumeration is infeasible, so ground truth is established by combining manual analysis with targeted simulation (1000 random schedules for each candidate race pair).

G.4 Time Breakdown

Table 4 provides a detailed breakdown of analysis time for each pipeline stage.

The dominant costs are trace collection and MCTS search, which together account for 85–96% of total time across all configurations. The abstract interpretation and decomposition stages are negligible ($< 2\%$ of total time), validating the efficiency of the constrained zonotope domain.

Table 4: Time breakdown by pipeline stage (seconds). Percentages show fraction of total time.

		Pipeline Stage						
Scenario	n	Trace	HB	Decomp	Abstract	MCTS	IS	**Total**
Highway	2	0.012	0.001	0.003	0.001	0.020	0.002	0.039
Highway	3	0.027	0.001	0.004	0.001	0.037	0.002	0.072
Highway	4	0.048	0.001	0.006	0.001	0.063	0.002	0.121
Warehouse	4	0.053	0.008	0.004	0.001	0.066	0.002	0.134
Warehouse	6	0.134	0.002	0.006	0.001	0.136	0.004	0.283
Warehouse	8	0.277	0.002	0.010	0.002	0.224	0.004	0.519

G.5 Zonotope Statistics

Table 5: Zonotope propagation statistics for the detection experiments.

Scenario	n	**HB Events**	**HB Edges**	**Coverage (%)**	**Abs. Error**
Highway	2	300	285	3.62	0.036
Highway	3	400	380	3.98	0.040
Highway	4	500	475	4.71	0.047
Warehouse	4	500	475	2.09	0.047
Warehouse	6	700	665	3.21	0.071
Warehouse	8	900	855	5.45	0.121

The HB graphs are dense (95% of event pairs are causally ordered), reflecting the tight physical coupling in both benchmarks. The coverage percentage (fraction of the state space covered by the zonotope) ranges from 2.09% to 5.45%, indicating that the abstract domain effectively focuses on the relevant portion of the state space. The abstraction error grows sub-linearly with agent count, consistent with the theoretical bound from Section D.

H Zonotope Reduction

As the analysis propagates through ReLU layers, the number of zonotope generators grows. To maintain computational tractability, we periodically reduce the generator count.

H.1 Reduction Strategy

Given a zonotope $Z = \mathsf{Zono}(c, G)$ with $p > p_{\max}$ generators, we reduce to $p_{\max}$ generators using the following procedure:

1. **Sort generators by norm:** order $g_1, \ldots, g_p$ such that $\|g_1\| \geq \|g_2\| \geq \cdots \geq \|g_p\|$.
2. **Keep top generators:** retain $g_1, \ldots, g_{p_{\max}-d}$.
3. **Merge remaining generators:** replace $g_{p_{\max}-d+1}, \ldots, g_p$ with d axis-aligned generators forming the interval hull:

$$g'_k = \left(\sum_{j=p_{\max}-d+1}^{p} |g_{jk}| \right) e_k, \quad k = 1, \ldots, d.$$

H.2 Bounded Error Analysis

Theorem H.1 (Reduction error bound)**.** *Let $Z = \mathsf{Zono}(c, G)$ be a zonotope and $Z' = \mathsf{Zono}(c, G')$ the reduced zonotope obtained by the above procedure. Then:*

$$Z \subseteq Z' \subseteq Z \oplus B_\infty(\epsilon_{red})$$

where $\epsilon_{red} = \max_k \sum_{j=p_{\max}-d+1}^{p} |g_{jk}|$ and $B_\infty(\epsilon)$ is the ℓ_∞-ball of radius ϵ.

Proof. Containment $Z \subseteq Z'$. For any $s = c + \sum_j \xi_j g_j \in Z$ with $\xi \in [-1, 1]^p$, we construct $\xi' \in [-1, 1]^{p_{\max}}$ such that $s = c + G'\xi'$. For the retained generators, $\xi'_j = \xi_j$ for $j \leq p_{\max} - d$. For the merged generators, $\xi'_{p_{\max}-d+k} = \sum_{j>p_{\max}-d} \xi_j g_{jk}/g'_k$ for each axis k. By the triangle inequality:

$$|\xi'_{p_{\max}-d+k}| = \frac{|[|]\sum_j \xi_j g_{jk}}{\sum_j |g_{jk}|} \leq \frac{\sum_j |g_{jk}|}{\sum_j |g_{jk}|} = 1.$$

Hence $s \in Z'$.

Over-approximation bound. For any $s' = c + G'\xi' \in Z'$, we can write $s' = c + \sum_{j=1}^{p_{\max}-d} \xi'_j g_j + \sum_{k=1}^{d} \xi'_{p_{\max}-d+k} g'_k e_k$. The last d terms contribute at most $|g'_k|$ in each dimension k, while the corresponding terms in Z contribute at most $\sum_{j>p_{\max}-d} |g_{jk}| = g'_k$. The over-approximation error is thus exactly $\epsilon_{\text{red}} = \max_k g'_k$ per dimension. □

H.3 Adaptive Reduction

In practice, we use an adaptive reduction strategy:

- **Threshold:** Reduction is triggered when $p > 2p_{\max}$ to amortize the cost.
- **PCA-based merging:** Instead of axis-aligned merging, we compute the principal components of the small generators and merge along the top-d principal directions. This reduces the approximation error when the small generators are correlated.
- **Error tracking:** The cumulative reduction error is tracked and reported as part of the abstract error (Table 5).

Proposition H.2 (PCA merging improvement)**.** *PCA-based merging reduces the over-approximation volume by a factor of at most $\prod_{k=1}^{d} \sigma_k/\sigma'_k$ compared to axis-aligned merging, where σ_k and σ'_k are the singular values of the small generator matrix in the original and rotated coordinate systems, respectively.*

Proof. The volume of the over-approximation box in the axis-aligned case is $\prod_k 2\sum_j |g_{jk}|$. In the PCA-rotated coordinate system, the box volume is $\prod_k 2\sigma_k$ where σ_k are the singular values of $[g_{p_{\max}-d+1} \cdots g_p]$. Since the singular values satisfy $\sigma_k \leq \sum_j |g_{jk}|$ with equality only when all small generators are axis-aligned, PCA merging always yields equal or smaller over-approximation volume. □

For the configurations in our experiments, PCA merging was not necessary (the generator count stayed below $p_{\max} = 100$ throughout), but the mechanism is available for larger networks with more unstable neurons.